THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapters 3 & 4: **The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 17 u	14 + 3×1
2	**(b)** 100 u	40 + 12 + 48
3	**(a)** 8.8 g	$\frac{44}{12} \times 2.4 = 8.8$
4	**(b)** 12 g	$\frac{4}{3} \times 9 = 12$
5	**(c)** MgO	Mg^{2+} and O^{2-}
6	**(d)** 20	neutrons = 40 − 20
7	**(b)** 2, 8, 1	fill shells in order with maximum 2, 8, 8, 2
8	**(a)** 2, 8, 8, 2	fill shells in order with maximum 2, 8, 8, 2
9	**(b)** 1	electrons lost, gained or shared to complete the octet
10	**(c)** 32	$2n^2 = 2 \times 4^2 = 32$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Neutral entity of more than one atom capable of independent existence (1); H_2 / O_2 and H_2O / HCl (1).	1+1
12	By sharing of electron pairs between atoms to complete the octet/duplet (1); e.g. H_2, Cl_2 or HCl (1).	1+1
13	Atomic number = number of protons in the nucleus (1); mass number = protons + neutrons (1).	1+1
14	Protons = electrons = 26 (1); neutrons = 56 − 26 = 30 (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	O (2,6) needs two electrons; each H (1) needs one (1); O shares one electron with each of two H atoms (1); H_2O has two single bonds (1)—each atom becomes stable.	1+1+1
16	Z = 35 − 18 = 17, chlorine (1.5); 2, 8, 7 (1.5).	1½+1½
17	Na (2,8,1) loses 1 electron to become Na^+ (2,8) (1); Cl (2,8,7) gains 1 electron to become Cl^- (2,8,8) (1); opposite charges attract forming an ionic bond (1).	1+1+1
	OR: Mg (2,8,2) loses two electrons to form Mg^{2+} (1); each Cl gains one electron: two Cl^- (1); formula $MgCl_2$ — two chloride ions per magnesium ion (1).	1+1+1
18	Neutrons 30, electrons 26 (2), symbol Fe (1).	2+1
	OR: Na (2,8,1) loses 1 electron, valency 1 (1); O (2,6) gains 2, valency 2 (1); C (2,4) shares 4, valency 4 (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) NaCl solution (1). (ii) It forms free ions Na^+ and Cl^-; sugar does not form ions (2). (iii) NaCl has a high melting point; sugar has low (1). (iv) NaCl ionic; sugar covalent (1).	1+2+1+1
	OR: (i) $CaCl_2$, Al_2O_3, Na_2SO_4, MgO, NH_3 (2.5). (ii) Cross-over of valencies / criss-cross method (1). (iii) carbon monoxide, carbon dioxide, carbon disulfide, dinitrogen tetroxide, phosphorus trichloride (1.5).	2½+1+1½
20	(i) Atoms of the same element with the same atomic number but different mass numbers (1). (ii) Protium 1p, deuterium 1p+1n, tritium 1p+2n (2). (iii) Same number of electrons give similar chemistry; different masses change density, melting and boiling points (2).	1+2+2
	OR: (i) Chlorine-35 and chlorine-37 (1). (ii) (35 + 37)/2 = 36 u (1). (iii) 35 × 0.75 + 37 × 0.25 = 35.5 u (2). (iv) The isotopes are not present in equal amounts (1).	1+1+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET B

Chapters 3 & 4: **The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** phosphorus trichloride	prefixes show the number of atoms; second element ends in -ide
2	**(a)** Atoms cannot be created or destroyed in a chemical reaction	indivisible and conserved in reactions
3	**(d)** 3	N≡N triple bond (5 valence electrons each)
4	**(a)** $MgCl_2$	metal + non-metal; electron transfer
5	**(c)** sugar is covalent and does not form ions	no free charge carriers
6	**(b)** embedded in a sphere of positive charge	positive sphere with negative charges
7	**(a)** the stability of the atom	an orbiting electron would lose energy
8	**(d)** neutron	no charge
9	**(c)** isobars	e.g. ^{40}Ar and ^{40}Ca
10	**(b)** 35.5 u	(35 × 0.75) + (37 × 0.25)

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	(a) carbon monoxide (b) dinitrogen pentoxide (c) hydrogen sulfide (d) sulfur hexafluoride (½ each).	½+½+½+½
12	Na (2,8,1) loses one electron to form Na^+; Cl (2,8,7) gains it to form Cl^-; the ions attract (2).	1+1
13	Maximum electrons in a shell = $2n^2$ (K:2, L:8, M:18); outermost shell holds at most 8; shells fill in order K, L, M… (2).	1+1
14	Na: 2, 8, 1 (valency 1); Cl: 2, 8, 7 (valency 1) (1+1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Al^{3+}, O^{2-} → Al_2O_3 (1); $CaCO_3$ (1); Na_2SO_4 (1).	1+1+1
16	Isotopes: same Z, different A; e.g. ^{12}C and ^{14}C (1.5). Isobars: same A, different Z; e.g. ^{40}Ar, ^{40}K, ^{40}Ca (1.5).	1½+1½
17	carbon disulfide; dinitrogen tetroxide; phosphorus trichloride; hydrogen sulfide; ammonia (2). Prefixes mono, di, tri… show numbers of atoms; no prefix on hydrogen or mono for the first element (1).	2+1
	OR: (i) Transfer vs sharing of electrons (1); (ii) high vs low (1); (iii) conduct in molten/aqueous state vs generally non-conductors (1).	1+1+1
18	Protium (1p, 0n), deuterium (1p, 1n), tritium (1p, 2n) (1+1+1).	1+1+1
	OR: U-235: nuclear fuel; Co-60: cancer treatment; I-131: goitre; C-14: dating fossils (any three) (1+1+1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Bond: transfer vs sharing of electrons (1); state: crystalline solids vs gases/liquids/soft solids (1); melting point: high vs low (1); conductivity: conduct when molten/dissolved vs non-conductors (1). (ii) Water molecules stabilise the ions and separate them; covalent compounds have no ions (1).	4+1
	OR: (i) CO_2 and NH_3 covalent; NaCl and $MgCl_2$ ionic (1). (ii) 44 u and 17 u (1). (iii) 58.5 u and 95 u (1). (iv) Ionic solids are lattices of ions without discrete molecules, so the simplest formula is used (2).	1+1+1+2
20	(i) Same mass number, different atomic numbers, e.g. ^{40}Ca and ^{40}Ar (1). (ii) Ca 20, K 21, Ar 22 neutrons (2). (iii) Isotopes: same Z, different A; isobars: same A, different Z (2).	1+2+2
	OR: (i) U-235 fuel; Co-60 cancer therapy; I-131 goitre; C-14 age of fossils; Na-24 flow in pipes (any) (2.5). (ii) Only radioactive isotopes emit the required radiation (1). (iii) Homi Jehangir Bhabha / Raja Ramanna (1.5).	2½+1+1½

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET C

Chapters 3 & 4: **The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 98 u	2 + 32 + 64
2	**(a)** 95 u	24 + 71
3	**(d)** 1 : 8	Law of constant proportions
4	**(c)** Al_2O_3	Al^{3+} and O^{2-} cross over
5	**(d)** $CaCO_3$	Ca^{2+} and CO_3^{2-}
6	**(b)** 14	neutrons = 27 − 13
7	**(a)** 2, 8, 7	fill shells in order with maximum 2, 8, 8, 2
8	**(a)** 2, 8, 3	fill shells in order with maximum 2, 8, 8, 2
9	**(d)** 2	electrons lost, gained or shared to complete the octet
10	**(d)** parmanus	Vaisesika Sutras

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	(a) $CaCl_2$ (b) Al_2O_3 (1+1).	1+1
12	CO_2: $12 + 32 = 44$ u; NH_3: $14 + 3 = 17$ u (1+1).	1+1
13	The number of electrons gained, lost or shared by an atom to complete its octet (1); decided by the electrons in the valence shell (1).	1+1
14	Atoms of the same element with the same atomic number but different mass numbers (1); protium, deuterium, tritium (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	H_2SO_4: $2 + 32 + 64 = 98$ u (1.5); $CaCl_2$: $40 + 71 = 111$ u (1.5).	1½+1½
16	Weighted average $= 35 \times 0.75 + 37 \times 0.25$ (1) $= 26.25 + 9.25 = 35.5$ u (2).	1+2
17	NaCl forms free Na^+ and Cl^- ions in water which carry current (2); sugar is covalent and does not produce ions (1).	2+1
	OR: Strong electrostatic attraction between ions (1); regular lattice needs much energy to melt (1); displacement of layers brings like charges together causing brittleness (1).	1+1+1
18	$79 \times 0.497 + 81 \times 0.503 = 39.263 + 40.743 = 80.006 \approx 80.0$ u (3).	1+1+1
	OR: Natural chlorine is a mixture of isotopes in unequal proportions (1); the weighted average takes abundances into account (1); no single atom has mass 35.5 u (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Energy released when nuclei split or combine; used in electricity generation, medicine, research (2). (ii) Raja Ramanna (1). (iii) Heat from nuclear reactions makes steam that drives turbines (2).	2+1+2
	OR: (i) Zinc chloride and hydrogen gas (1). (ii) Hydrogen escaped (1). (iii) Use a closed flask with a balloon or stopper (2). (iv) Equal masses before and after (1).	1+1+2+1
20	(i) Argon (1). (ii) Na and Mg (1). (iii) Cl (1). (iv) $MgCl_2$ — Mg loses two electrons, each Cl gains one (2).	1+1+1+2
	OR: (i) Thomson: sphere of positive charge with electrons; Rutherford: central nucleus with orbiting electrons, stability problem; Bohr: fixed shells, stable (3). (ii) Discovery of electrons/radioactivity (1). (iii) Thomson, Rutherford, Chadwick (1).	3+1+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET D

Chapters 3 & 4: **The Atomic Foundations of Matter, Journey Inside the Atom** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** sulfur hexafluoride	prefixes show the number of atoms; second element ends in -ide
2	**(a)** an electrically neutral entity of more than one atom capable of independent existence	definition
3	**(d)** loses one electron to form Na^+	completes octet by losing its single valence electron
4	**(d)** high melting and boiling points	strong attraction between ions
5	**(a)** ionic compounds	no discrete molecules in ionic solids
6	**(a)** most of the atom is empty space	Rutherford's inference
7	**(b)** revolve in fixed shells without losing energy	stationary states
8	**(a)** protons in its nucleus	defines the element
9	**(c)** iodine-131	radioactive iodine
10	**(d)** unreactive and stable	complete valence shell

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Molecular mass is for covalent compounds made of discrete molecules (1); formula unit mass is for ionic compounds, using the simplest formula (1).	1+1
12	Strong electrostatic force between oppositely charged ions (1); they form a regular crystal lattice needing much energy to break (1).	1+1
13	They have the same number of electrons and the same electronic configuration (2).	2
14	Atoms of different elements with the same mass number but different atomic numbers (1); ^{40}Ar and ^{40}Ca (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Their valence shell is complete (8 electrons or duplet for helium) so they are stable and do not combine (2); helium, neon, argon (1).	2+1
16	Their valence shells are complete (octet, or duplet for helium) (1); Ne 2, 8; Ar 2, 8, 8 (2).	1+2
17	When atoms combine the energy of the system becomes lower than that of the separate atoms (2), so the arrangement is more stable (1).	2+1
	OR: When hydrogen is the first element no prefix is used, so H_2S is hydrogen sulfide (1.5); H_2O is water and NH_3 is ammonia — known by common names (1.5).	1½+1½
18	Atomic number 11 for both; mass numbers 23 and 24 (2); same element — they are isotopes (1).	2+1
	OR: In the modern model electrons exist as electron clouds; only the regions of highest probability can be predicted (3).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) A neutral entity of more than one atom that can exist independently (1). (ii) Element: H_2, O_2, Cl_2; compound: H_2O, HCl, CO_2 (3). (iii) Its valence shell is already full (1).	1+3+1
	OR: (i) 12 : 32 = 3 : 8 (1). (ii) $\frac{12}{44} \times 22 = 6$ g (2). (iii) $\frac{8}{3} \times 6 = 16$ g (2).	1+2+2
20	(i) A standard international shorthand for communication (1). (ii) First letter capital, second letter small; many from first one or two letters; some from Latin, Greek or German names (2). (iii) Fe (ferrum), Hg (hydrargyros), Na (natrium), K (kalium) (2).	1+2+2
	OR: (i) Cathode rays were identical for every cathode material and gas, so electrons are a common constituent (1). (ii) Number of protons equals number of electrons, so the charges cancel (1). (iii) 13 protons, 13 electrons, 14 neutrons (1.5); configuration 2, 8, 3 (1); valency 3 (0.5).	1+1+3

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND